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Beyond Asking the Right Questions: Accounting for Error and Bias in a Recreational Fishery Survey

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ABSTRACT

Accurate information about anglers' motivations, satisfaction, and fishing behavior is needed to design effective recreational fisheries management. This study used a comprehensive survey of Swedish anglers to explore the risk for common sampling strategies and survey response groupings to introduce significant error or biases that impact the interpretation of data. Angler motivation and satisfaction differed between anglers based on avidity, type of boat use, and gender. Measures of satisfaction from angling appear to be less affected by biases than general motivations, fishery specific motivations, and catch measures. Fishing season and place of residence have less relationship to psychological dimensions of survey responses, but remain important when considering target species. These results can guide data collection to more accurately measure the preferences and behaviors of different resource user groups, improve the design of regulations and management, and tailor communication to facilitate greater compliance and adoption of best practices.

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Angler behavior; fishing motivation; fishing satisfaction; human dimensions; multispecies fishery; selection bias

Introduction

The sustainable use of natural resources requires evidence-based management supported by reliable science. Recreational fisheries research, to this end, often involves conducting surveys of anglers (people fishing with handheld hook and line tackle), which simultaneously captures information about both the biological and the human dimensions of the resource use system (Nieman et al. 2021). For example, surveys are used to collect data about recreational catches (e.g., catch-per-unit-effort; CPUE) (Ferter et al. 2023), and to better understand anglers' motivations to fish (Beardmore et al. 2011), satisfaction with the fishing experience (Birdsong et al. 2022), attitudes toward

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management (Curtis et al. 2019; van den Heuvel et al. 2020), the value of recreational fishing to anglers (Blicharska and Rönnbäck 2018) and other human dimensions. Through proper survey design, researchers aim to accurately capture these dimensions and minimize the four main types of error (coverage, sampling, non-response, and measurement) that contribute to differences between survey data and the true values of the variables that they represent (Dillman, Smyth, and Christian 2014). The introduction of these errors at various stages of the research process can contribute to biases that shift sample statistics in one direction or another. These biases could lead researchers and fishery managers to draw the wrong conclusions about the recreational fishery and angler population, which could result in certain issues being overlooked or poor management decisions being made, ultimately undermining the actual needs of the fishery and fish populations (Taylor et al. 2025). Although generating some error is inherent when taking a sample rather than surveying an entire population, understanding the characteristics of the target population can guide the careful selection of sampling frames and recruitment methods that reduce error and subsequent biases.

Many conventional angler survey methods (NRC 2006) are sensitive to biases due to their sampling frames introducing issues such as coverage or sampling errors. For example, location-based sampling can influence harvest rates data (Dorow and Arlinghaus 2011); survey timing may obscure diel and seasonal variation in fishing (Hartill and Edwards 2015; Sande et al. 2022); and survey focus on an individual species or fishery can emphasize responses relating to particular angler motivations (Beardmore et al. 2011). In addition, cluster sampling, for example through mailing lists for fishing clubs or associations (Jones et al. 2025; van den Heuvel et al. 2020), may introduce an avidity bias (Thomson 1991). Anglers in formal organizations tend to be more avid, active, and experienced than anglers that are not organized in this way (Lewin, Weltersbach, Haase, Riepe, et al. 2021; Oh et al. 2005; Schramm and Gerard 2004), which has been shown to influence numerous human dimensions (Hutt and Bettoli 2007; Hyman and McMullin 2018). Unit non-response error (failure to receive a valid response from a sampled unit) is a notable risk with regard to gender in recreational fishing studies as demonstrated by Duda and Nobile (2010) where the percentage of females responding to their survey was less than half of the percentage of females in the entire sample population. In the few studies that do take gender into account, the experiences of female anglers can be quite different from those of males (Burkett and Carter 2022; Fennell and Birbeck 2019; Floyd et al. 2006), but the implications of this underrepresentation to research and management remain unclear.

Efforts have been made to address some of these challenges through various approaches (e.g., weighting, drop-out removal, proportionate or disproportionate sampling, multi-frame sampling) to compensate for error and biases in angler survey data (Dorow and Arlinghaus 2011; Sande et al. 2022; Taylor et al. 2025; van der Hammen, de Graaf, and Lyle 2016). However, weighting only guarantees that a sample is representative with respect to characteristics that are correlated with the one used for weighting so that some estimates may be improved, some unaffected, and introducing biases to others (Dillman, Smyth, and Christian 2014). Recognizing the diversity of angler populations (Hunt et al. 2023) and acknowledging how the selection of respondents influences survey results (Duda and Nobile 2010; Gundelund et al. 2023; Lewin et al. 2023) helps researchers to anticipate and potentially compensate for errors and

biases in survey data. Furthermore, many studies focus on only a handful of potential causes and effects for differences among respondent groups and make limited efforts to evaluate the broad scope of factors that may introduce error or bias and the relative impacts of each. However, a great deal more information and understanding are needed around the human dimensions of recreational fisheries, especially as the focus has so far been mostly on socio-demographic and catch-related human dimensions rather than psychological dimensions of recreational fishing, such as motivations to fish and satisfaction (NRC 2006).

The aim of the current study conducted in Sweden was to identify whether samples of the same population generated by means of different commonly used recruitment strategies led to different outcomes. We hypothesized that actual sampling frames based on membership in the Swedish national angling association (Sportfiskarna) or general public would result in different survey responses regarding catch-related variables (e.g., catch per unit effort (CPUE) and release rates) and psychological dimensions of fishing (e.g., motivations to fish and angler satisfaction). We also hypothesized that correlated recruitment strategy proxies (e.g., boat use, gender, and organization membership) would result in consistently similar survey responses across catch-related and psychological dimension questions. Based on the results of our analyses, we discuss how different recruitment strategies should be considered during the design and interpretation of angler surveys, with particular emphasis on the collection of much needed data on the psychological dimensions of recreational fisheries.

Methods

Study Area and Recruitment

The Roslagen archipelago encompasses the Baltic Sea coastlines of Stockholm County and much of Uppsala County in Sweden. This study defines the archipelago as stretching from the north boundary of Östhammar municipality to south boundary of Stockholm County (Figure 1), following previous studies of fishing activity in the area (Söderqvist et al. 2005; Soutukorva and Söderqvist 2005). This delineation focuses the study on a relatively homogenous geographical area with uniform fishing regulations (HVMFS 2023).

Recreational angling in Sweden's coastal waters and five largest lakes does not require a license. In addition to international agreements (EU Common Fisheries Policy) national regulations are set by the Swedish Agency for Marine and Water Management, and County governments (SvenskaFiskeregler 2023). The brackish waters of the archipelago are host to a mix of freshwater (e.g., northern pike (*Esox lucius*), and European perch (*Perca fluviatilis*)), marine (e.g., Atlantic cod (*Gadus morhua*), and Atlantic herring (*Clupea harengus*)), and anadromous species (e.g., sea trout (*Salmo trutta*), and Atlantic salmon (*Salmo salar*)).

In 2018, the population of anglers in the study area was surveyed through two sampling frames: 1) the general public, and 2) members of Sportfiskarna, which is a national organization representing the interests of recreational anglers with the primary goal of ensuring healthy fish stocks and inland and coastal waters. Both samples were limited to individuals residing in the target counties (i.e., Uppsala and Stockholm) of

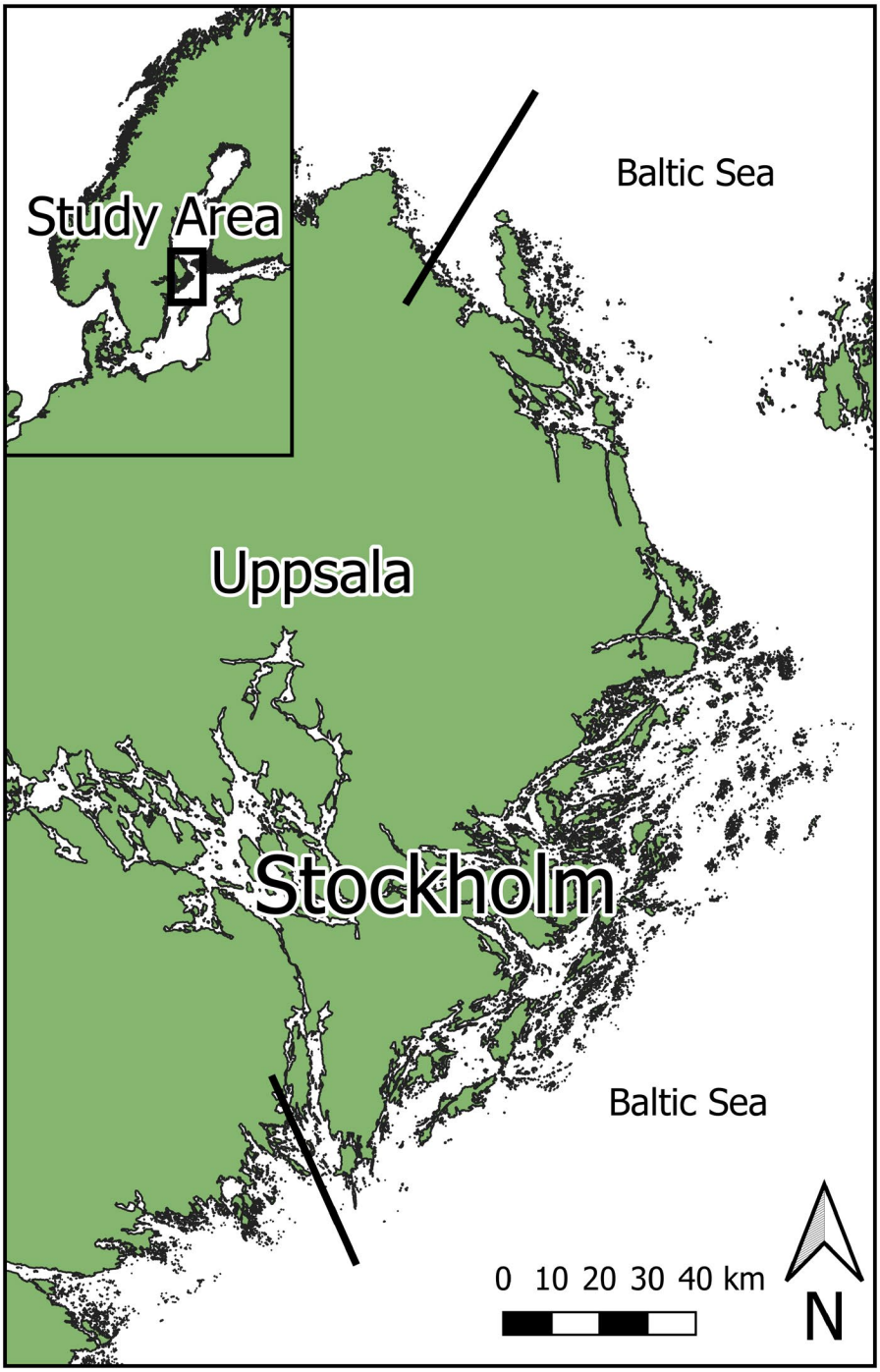


Figure 1. Location of the Roslagen Archipelago around Stockholm, Sweden. Black lines denote the north and south boundaries of the study area.

which the Roslagen archipelago is a part. For the general public, a list of physical addresses for 2000 people aged 16–75 years was purchased from SPAR (the state's personal address register). A total of 1500 Sportfiskarna members were contacted

through randomly selected e-mail addresses from the association's membership register in the target counties. At the time of the study, Sportfiskarna's membership in Stockholm and Uppsala counties included a total of 7,354 people, of which 5,417 had provided an e-mail address. Gender and age distribution were similar for Sportfiskarna members regardless of whether e-mail was provided or not, therefore the members targeted via email were assumed representative of this study population. At the end of 2018, the population aged 16–75 in Stockholm and Uppsala County was 2,012,858 people, according to Statistics Sweden's database (compared to 1,984,698 in 2017).

The members of the general public within our sampling frame were sent a letter by post with a description of the study and a short weblink with an associated QR code to the survey website (SurveyMonkey). Members of Sportfiskarna were emailed the corresponding description with a digital link to the same website. No personal login was required to participate in the survey, and answers were completely anonymous. In Sweden this type of survey and the questions asked do not require an ethics review.

The survey was conducted after the summer season (June–August) of 2018. Additionally, Sportfiskarna members were also surveyed after the spring (March–May) and autumn (September–November), contacting 500 members for each season. In addition to the initial invitation to the survey, two reminders were sent approximately ten days apart. The questionnaire was answered completely anonymously, but in order to manage reminders to the public, respondents who finished the survey were given the opportunity to fill in their name and zip code at the end. This information was used solely to sort out the reminder mailings. For Sportfiskarna, the reminders were handled digitally, which was notably more efficient and less expensive than sending reminders by mail.

Survey Design

The survey design largely followed previous studies of fishing activity in the archipelago (Söderqvist et al. 2005; Soutukorva and Söderqvist 2005), but also drew inspiration from national angling surveys (HaV 2019) to enable national-regional comparisons. In addition to the recruitment strategy proxies, catch-related and psychological variables described below, the survey included questions about the respondents' fishing activity (within and outside the archipelago), boat use and ownership in the archipelago, and residence or property ownership in the archipelago. Based on the survey questions, specific characteristics of the survey respondents were used to group them into categories (Figure 2) that reflected potential sources of error or bias.

Recruitment Strategy Proxies

First, avidity, often associated with potential oversampling of avid anglers (Thomson 1991) and which can relate to both sampling bias and non-response bias, was assessed by comparing the general public and Sportfiskarna sample frames. The assumption that these represented different ends of the avidity spectrum (i.e., low mean avidity among the general public versus high mean avidity among members of Sportfiskarna) was validated statistically (see "Analysis" for details). Then, the datasets from both sample frames were merged and recruitment strategy proxies were made by grouping

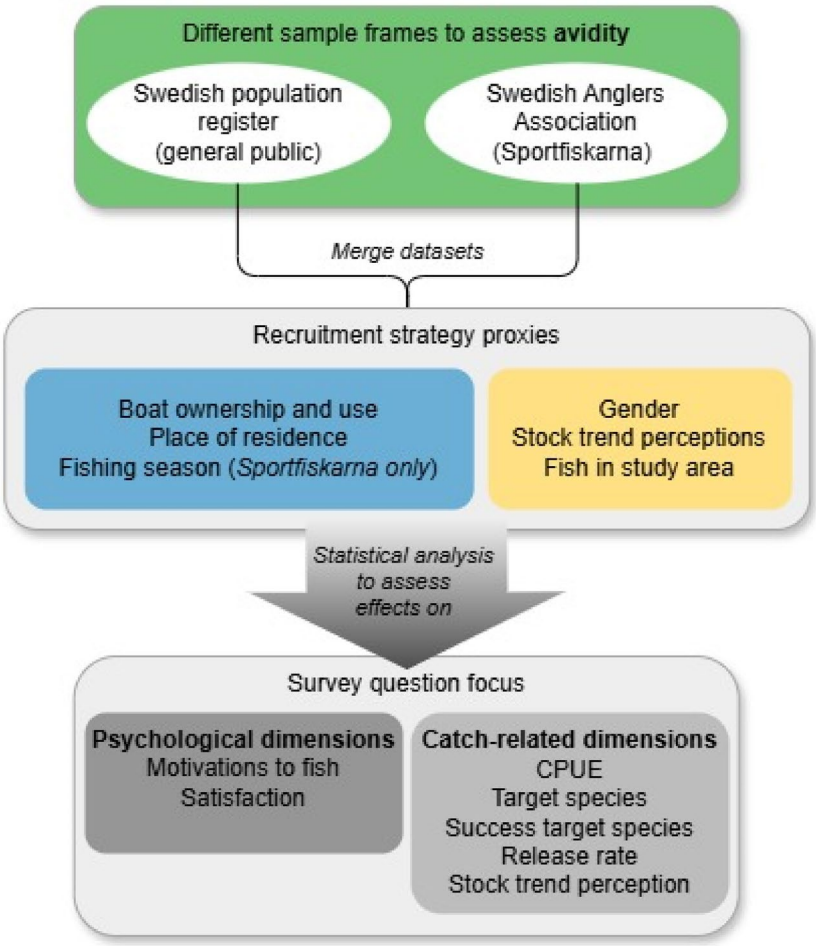


Figure 2. Recruitment strategy proxies and survey question focuses included in the study.

responses based on characteristics that represented common sampling methods in recreational fisheries research (see Table 1 for specific proxy descriptions and which recruitment strategies they reflect). Boat ownership and use for fishing were used as a proxy for boat ramp intercept surveys, and whether or not they fished in the archipelago during the study period (only used when comparing motivations for fishing in general) as a proxy for other intercept surveys. Sampling season was used to represent survey timing and targeting of seasonal fisheries. Permanent or temporary residence in the archipelago provided a proxy for narrower address-based sampling. Respondent gender provided an avenue to explore issues of nonresponse error and disproportionate sampling. Stock trend perception was used as a proxy for the circumstance or context of surveys focusing on fisheries with threatened or imperiled stocks. Several of these recruitment strategy proxies were also expected to be correlated with angler avidity. For example, boat ownership and use may also indicate the level of personal investment in fishing, which is expected to be higher in more avid anglers (Beardmore et al. 2011).

Table 1. The distribution of respondents among recruitment strategy proxies with descriptions of the defining characteristics of each.

Recruitment strategy proxy	Sections	Number	Description	Represented Strategy
Initial Sampling Frames	General Public	229	General public fishing during the summer 1 Jun-31 August	Address based sampling,
	Pooled Sportfiskarna	593	Sportfiskarna members pooled across spring, summer, and autumn samples	Snowball sampling
Boat Ownership and Use for Fishing	Own and Use	213	Respondent or someone in their family owns a boat, and use a boat for fishing in the archipelago	Boat ramp intercept survey
	Use Only	35	Use a boat for fishing in the archipelago, but do not own a boat	
	Own Only	203	Respondent or someone in their family owns a boat, but do not use a boat for fishing	
	None	371	Do not own or use a boat for fishing	
Gender	Male	598		Disproportionate sampling
	Female	93		
Residence in Roslagen Archipelago	Permanent	97	Respondent lives permanently in the archipelago	Address-based sampling
	Temporary	171	Respondent or someone in their family owns a summer house or fishing rights in the archipelago, but does not live there permanently	
	Nonresident	554	Respondent does not have permanent or temporary residence in the archipelago	
Seasons	Spring	159	Sportfiskarna members fishing during the spring 1 March-31 May	Seasonal timing of surveys
	Summer	233	Sportfiskarna members fishing during the summer 1 Jun-31 August	
	Fall	201	Sportfiskarna members fishing during the autumn 1 September-30 November	
Median perceived stock trend for target species[a]	Much worse	158	Perceived median trend that stocks are much worse for species targeted in the archipelago	Surveys focused on threatened or imperiled stocks
	Worse	234	Perceived median trend that stocks are worse for species targeted in the archipelago	
	Unchanged	99	Perceived median trend that stocks are unchanged for species targeted in the archipelago	
	Better	16	Perceived median trend that stocks are better for species targeted in the archipelago	
Fished in Archipelago	No	263	Respondent did not fish in the Roslagen Archipelago during the study period	Intercept surveys
	Yes	366	Respondent did fish in the Roslagen Archipelago during the study period	

Number refers to the number of usable responses for each category; the totals vary among recruitment strategy proxies. Represented strategy gives examples of commonly used recruitment strategies associated with each proxy.
[a]None of the respondents chose the option "Much better".

The respondents' perception of target species stock trend over the years that they have fished was assessed for each of the species that they have targeted in the archipelago, rating its status on a 5-point scale from *Much worse* to *Much better*. The median score for each respondent's stock trend perceptions across target species was used for further analysis to reflect the respondents' overall trend perception.

Catch-Related and Psychological Dimensions of Angling

Catch-related dimensions were assessed through CPUE, release rates, target species, and success at catching target species for the three latest fishing trips to the archipelago, building on variables explored by Soutukorva and Söderqvist (2005), Beardmore et al. (2011), and Lewin, Weltersbach, Haase, and Strehlow (2021). For each respondent the CPUE was calculated as the average number of fish caught across their reported trips. Release rates were calculated by taking the total number of fish released and dividing by the total number of fish caught across all trips for each respondent. Respondents were recorded as targeting a species if they indicated that it was a target on at least one of their reported fishing trips. The count of respondents indicating that they targeted a given species at least once was then used to compare the proportional targeting of species among recruitment strategy proxies. For each respondent overall success at catching targeted species was used as an indicator of relative angling skill and success in a fishery where multiple species may be targeted in a single trip and "good" catch rates may be highly variable among species. This was calculated across targets by first dividing the number of trips where at least one of a targeted species was caught by the number of trips where that species was targeted, then taking the average of these species-specific success rates across all of the species pursued by each respondent. These values were used as an indicator of relative angling skill and success in a fishery where multiple species may be targeted in a single trip and "good" catch rates may be highly variable among species. Counts of using individual fishing gear types on at least one fishing trip were used to compare patterns of gear use among respondent groups. Additionally, the respondents' median perceived stock trend (as described above) was explored as a dependent variable relative to the other recruitment strategy proxies.

Psychological dimensions measured were motivations to fish (within the archipelago and in general) and satisfaction with fishing in the archipelago, based on the questions used by Soutukorva and Söderqvist (2005) and Söderqvist et al. (2005). General motivations to fish and satisfaction with different aspects of fishing in the archipelago were measured on a 5-point agreement scale. In addition, respondents were asked to rank seven pre-determined motivations for fishing in the archipelago from most to least important.

Analysis

Analysis was limited to respondents that indicated that they had fished in Sweden during the last five years (general public n = 229, Sportfiskarna n = 593), and included respondents that fished in Sweden, but not in the archipelago during the study period.

Statistical analyses were performed using R version 4.2.1 (R Core Development Team 2020).

To validate the avidity differences between the public and Sportfiskarna members, non-parametric one-way permutation tests were performed, using the *coin* package in R (Hothorn et al. 2008). Permutation tests were appropriate in this case to compensate for varying sample sizes and non-equal variance across recruitment strategy proxies (Pesarin and Salmaso 2010). Avidity differences were validated by comparing the two samples in terms of the years of fishing experience and the number of days spent fishing in the study period (in general and in the archipelago). To assess the potential effect of avidity on the other recruitment strategy proxies included in this study, the public and Sportfiskarna samples were compared in terms of gender, place of residence, boat use and ownership, and stock trend perceptions. The fishing season was not included in this comparison, because the general public were only sampled in one season (i.e., summer). Demographic details for both samples were presented to aid the interpretation of sample comparisons. Significant permutation tests were followed with post-hoc pairwise permutation tests using Benjamini-Hochberg adjusted p-values to control the false discovery rate (Mangiafico 2015). Fisher's exact tests were used to compare the proportions of respondents using boats and residing in the study area between the two samples. For all tests the significance level was set at $p < 0.05$. In addition, Spearman rank correlations were used to check for relationships among the various characteristics used to define recruitment strategy proxies.

Potential relationships between the recruitment strategy proxies and target species were explored through Fisher's exact tests followed by pairwise tests of independence. In this case, the distributions of fishing attention across the range of target species were compared among groups of anglers for each recruitment strategy proxy.

The effect of each of the recruitment strategy proxies on the catch-related and psychological variables discussed in the previous section was assessed via non-parametric one-way permutation tests. The permutation procedure used was similar to the one described above.

Results

Comparing the Initial Sampling Frames

Response rates ranged between 28% and 48% for the study populations and survey periods. The general public accounted for the lowest response rate, while Sportfiskarna members angling during the summer had the highest. Of the 2,000 letters sent to the public, 52 were returned to the sender, so the response rate was calculated for 1,948 mailings. Regarding the e-mail, the letter was opened by 400 people for the summer, 305 for the spring and 374 for the autumn (response rates were still calculated in relation to the number of mailings). Of those who answered the survey, over 80% completed it. In comparison to our sample frames the respondents skewed toward male with 35.2% female respondents vs 50.4% actual for the general public, and 3.5% female respondents vs 13.8% female members for Sportfiskarna. Respondent age distributions were representative of the averages for both groups.

The public and Sportfiskarna samples did not differ significantly in the average age of respondents, the distribution of permanent or temporary residence in the archipelago, and the proportion of fishing days spent in the archipelago (Table 2). Differences existed in the years of fishing experience and number of days spent fishing in the previous three months, which were significantly higher for Sportfiskarna members, for both general fishing and fishing in the study area. Sportfiskarna members had a significantly higher rate of success in catching their target species, and a higher overall CPUE. Boat use for fishing was also significantly higher for Sportfiskarna members than for respondents from the public. The distribution of fishing effort by gear type varied significantly between the general public and Sportfiskarna members ($p < 0.001$), with the fishers from the general public being more likely to limit their gear choices to spin and passive bait fishing. Sportfiskarna members reported using a wider range of gear types and fishing styles, which also included more frequent use of fly-fishing, trolling, and vertical fishing. The differences between these two samples were also supported by significant correlations among the majority of their demographic characteristics and the factors used to compare their fishing avidity in Table 2.

Comparing Recruitment Strategy Proxies Based on Catch Related and Psychological Dimensions

The recruitment strategy proxies presented a diverse range of influences on motivation, satisfaction, and catch-related dimensions, as summarized in Tables 3 and 4. Sportfiskarna membership, as well as the boat use and gender proxies, influenced many more responses than the residence, season, and perceived trends proxies. In general, responses to questions on angler satisfaction from fishing in the archipelago were less affected by any of the proxies than responses to the motivation questions or in catches.

The various recruitment strategy proxies revealed significant differences in the distribution of fishing effort across the range of target species present in the Roslagen archipelago (Table 3). This distribution was different between the general public and Sportfiskarna respondents. Seasonality showed a shift in target from sea trout in the spring and fall to perch in the summer. Within the Sportfiskarna responses significant seasonal differences were also noted in gear use. Temporary resident and female respondents' targeted zander (*Sander lucioperca)* and sea trout less than male, permanent, or nonresident respondents. Female respondents also indicated much higher targeting of other or unspecified species during fishing trips. Boat use correlated with more targeting of perch, pike, and zander, and less targeting of sea trout. Perceived stock trend was the only proxy with no significant relationship to the distribution of target species.

Catch related motivations for fishing in general and in the archipelago were both found to vary significantly across the public-Sportfiskarna, boat use, and gender proxies, as summarized in Table 4 (more detailed results are presented in Supplementary Table 1). However, when considering satisfaction from fishing in the archipelago there were only limited significant differences and low test-statistic scores for the public-Sportfiskarna, boat use, and gender proxies, and no significant differences in catch motivation or satisfaction when comparing proxies based on

Table 2. Comparison of general demographics and key characteristics relating to fishing avidity between Sportfiskarna members and the general public.

			Public	Sportfiskarna	Significance
	Age	mean±sd	47.3 ± 16.4	47.7 ± 13.1	Z = −0.33,
		n	209	477	$p = 0.74$
Demographic	Income	mean±sd	27648 ± 14238	32457 ± 11939	Z = −4.45,
		n	202	470	$p < 0.001$
	Gender	female	35.2% (n = 74)	3.5% (n = 17)	$p < 0.001$
		male	64.8% (n = 136)	96.5% (n = 462)	
Residence in the archipelago	Permanent		9.6% (n = 22)	12.7% (n = 75)	$p = 0.423$
	Temporary		18.8% (n = 43)	21.6% (n = 128)	
	None		71.6% (n = 164)	65.8% (n = 390)	
General	Years Experience	mean±sd	24.4 ± 20.8	35.0 ± 15.5	Z = −6.89,
		n	189	481	$p < 0.001$
	Days Fishing	mean±sd	2.5 ± 6.5	11.0 ± 13.8	Z = −8.52,
		n	225	584	$p < 0.001$
Archipelago	Years Experience	mean±sd	11.8 ± 17.4	21.3 ± 16.3	Z = −6.40,
		n	187	466	$p < 0.001$
	Days Fishing	mean±sd	3.1 ± 4.9	7.7 ± 10.5	Z = −2.80,
		n	42	273	p = 0.0052
Proportion of fishing days spent in the archipelago		mean±sd	0.3 ± 0.5	0.4 ± 0.4	Z = −1.11,
		n	113	504	$p = 0.27$
Success	CPUE	mean±sd	2.7 ± 6.0	5.6 ± 9.9	Z = −1.91,
		n	45	321	$p = 0.056$
	Catch Target Species	mean±sd	0.3 ± 0.4	0.5 ± 0.4	Z = −3.35,
		n	40	283	$p < 0.001$
Perceived stock trend		mean±sd	−0.9 ± 0.8	−1.0 ± 0.6	Z = 0.77,
		n	90	417	$p = 0.444$
Boat	Proportion of trips fishing from boat	mean±sd	0.4 ± 0.5	0.6 ± 0.4	Z = −2.99,
		n	45	321	$p = 0.0028$
	Own boat and use for fishing		7.0% (n = 16)	33.2% (n = 197)	$p < 0.001$
	Use boat for fishing but do not own		1.3% (n = 3)	5.4% (n = 32)	
	Own a boat but do not use for fishing		21.4% (n = 49)	26.0% (n = 154)	
	Do not own or use a boat for fishing		70.3% (n = 161)	35.4% (n = 210)	

season, place of residence, and perceived stock trends. Motivations relating to the opportunity to catch fish, and catch big fish were scored significantly higher by respondents that fished from boats, by members of Sportfiskarna, and by those who identified themselves as male. At the same time, the ability to keep catches was scored lower as a motivation in these proxies, with the exception of those who fished from boats, but did not have access to their own. The patterns in motivation scores across these proxies were also reflected in their reported CPUE, success at catching target species, and release rates. However, there was significantly higher CPUE and success at catching target species for respondents that fish with boats that they own.

Scoring of responses related to the nature experience while fishing had contrasting influences across recruitment strategy proxies, and across general motivations, archipelago specific motivations and satisfaction from angling. Respondents that used boats scored the nature experience as a less important motivation for general fishing when compared to respondents not using boats, but this difference was not present when considering their satisfaction from fishing in the archipelago. The role of nature was not significantly different between the general public and Sportfiskarna members when

Table 3. Percentage of respondents targeting different species.

		Target species								
Proxy	Pairwise Grouping	Perch	Pike	Zander	Sea Trout	Salmon	Whitefish	Herring	Cod	Other
Study Population	$p < 0.001$									
Public	a	69.2	51.3	7.7	9.0	1.3	2.6	7.7	1.3	7.7
Sportfiskarna	b	46.3	58.4	15.6	23.1	2.3	0.6	3.6	0.9	1.1
Spring	b	39.0	59.1	17.0	35.8	3.1	0.0	5.0	0.6	1.3
Summer	c	59.2	60.5	16.7	12.2	3.1	0.0	4.4	1.0	2.0
Fall	b	35.7	55.3	13.5	27.9	0.8	1.6	1.6	0.8	0.0
Boat Use	$p < 0.001$									
Own and use	a	49.1	66.3	16.2	17.3	2.1	0.8	3.9	0.8	1.2
Use only	ab	58.2	59.7	20.9	19.4	7.5	1.5	7.5	3.0	1.5
Own only	bc	44.6	35.6	9.9	25.7	0.0	0.0	2.0	1.0	4.0
None	c	41.2	31.8	8.2	45.9	1.2	1.2	4.7	0.0	3.5
Gender	$p < 0.001$									
Female	a	58.1	35.5	16.1	3.2	0.0	6.5	9.7	0.0	19.4
Male	b	52.1	63.1	14.8	24.2	2.7	0.6	4.1	1.1	1.3
Residence	$p = 0.004$									
Permanent	a	52.7	53.3	21.9	20.7	1.8	0.0	5.3	1.2	3.0
Temporary	b	50.0	60.6	7.6	14.4	3.4	0.8	3.8	0.4	1.7
Nonresident	a	45.5	58.3	16.3	27.0	1.6	1.1	3.5	1.1	1.4
Perceived Trend	$p = 0.1$									
Much worse	a	45.4	62.1	14.6	29.2	3.8	0.4	4.6	0.4	2.1
Worse	a	59.1	63.1	17.4	17.8	2.3	1.7	5.7	1.7	0.7
Same	a	50.0	71.1	14.5	23.7	0.0	0.0	1.3	1.3	0.0
Better	a	66.7	50.0	10.0	20.0	0.0	0.0	0.0	0.0	3.3

Total percentages can equal more than 100 as it was possible to report targeting multiple species per trip. Similarity groupings based on pairwise tests of independence using adjusted *p*-values are indicated for each proxy.

considering fishing motivation, but received a significantly higher score from Sportfiskarna members for its role in satisfaction from fishing in the archipelago.

When compared to the other dependent variables, socialization and relaxation as motivations for fishing and measures of satisfaction both presented unique response patterns that contradict some of the broad correlations that were present across the recruitment strategy proxies. In motivations for fishing in general and fishing in the archipelago, social interaction with family and friends was scored as significantly more important by members of the general public and females than by Sportfiskarna members and male respondents. However, these differences are not present within other recruitment strategy proxies when comparing satisfaction from social factors and the pursuit of solitude when fishing in general. Similarly, the opportunity to relax was only given a higher score of importance in Sportfiskarna members' general motivations for fishing and in temporary residents' motivations to fish in the archipelago. Relaxation did not receive significantly different scores in motivation or satisfaction within any of the other proxies.

Whether or not respondents fished in the archipelago during the study period had significant relationships with the responses to six of the nine general motivation questions. Those who fished in the archipelago had significantly higher mean scores for the importance of the opportunities to catch fish and catch big fish. They also placed significantly lower scores on the importance of keeping their catches, the nature experience, accessibility, and solitude. No significant difference was found for the importance of information availability, social interaction with family and friends, or relaxation for

Table 4. Comparison of influences from selected recruitment strategy proxies on the mean responses to catch-related measures, general motivation to fish, motivation to fish in the archipelago, and satisfaction with fishing in the archipelago using values from the one-way analysis with permutation test of independence.

	Measure, motivation, or satisfaction	SF or Public[a]	Boat[b]	Gender[a]	Residence[b]	Seasons[b,c]	Perceived Trend[b]	Fished in Archipelago[a]
Catch-related management factors	Catch per unit effort	−1.91	5.78***	1.89	2.16	1.62	0.93	
	Success at catching target species	−3.35***	6.12***	2.66**	2.01	2.01	0.72	
	Release rate	−2.70**	2.98*	1.26	0.90	1.32	1.86	
	Perceived stock trend	0.76	2.87*	1.90	1.09	1.43		0.60
What importance do the following aspects have when you decide to fish in general?	Opportunity to catch fish	−5.87***	4.23***	3.62***	0.71	0.78	1.45	−2.63**
	Opportunity to catch big fish	−7.82***	5.36***	5.70***	1.48	1.58	1.10	−3.28**
	Ability to keep caught fish	5.81***	4.07***	−3.68***	1.35	0.79	1.68	2.11*
	The nature experience	−0.93	2.63*	−0.29	2.83*	0.54	1.65	2.69**
	Accessibility	0.60	3.68***	−2.49*	2.20	1.58	1.62	3.34***
	Being alone at the fishing spot	−1.55	0.81	1.37	2.27	0.42	1.24	2.62**
	Information on fishing rules	−3.45***	1.79	0.76	1.07	1.59	1.36	1.67
	Socializing with family and friends	2.23*	2.16	−2.28*	0.73	1.43	1.72	−0.19
	Opportunity to relax	−2.66**	0.70	0.57	0.56	1.58	1.83	0.55
Ranking importance of reasons to fish in the archipelago	Opportunity to catch fish	−6.43***	4.12***	3.68***	0.84	0.69	1.08	
	Fish are food	4.11***	2.85*	−2.40*	1.27	1.00	0.60	
	Opportunity to be in nature	1.56	2.37	−2.58**	0.64	1.56	0.60	
	Fishing with hand gear is free	−2.09*	2.11	2.53*	2.65*	1.74	1.53	
	Socializing with family and friends	4.92***	2.60	−3.00**	1.51	0.50	1.53	
	Opportunity to relax	−0.25	1.67	0.72	2.44*	0.52	1.19	
	Other	−0.46	2.44	0.12	1.46	0.63	1.92	

(*Continued*)

Table 4. Continued.

	Measure, motivation, or satisfaction	SF or Public[a]	Boat[b]	Gender[a]	Residence[b]	Seasons[b,c]	Perceived Trend[b]	Fished in Archipelago[a]
To what extent were these aspects satisfied when fishing in the archipelago?	Opportunity to catch fish	−1.93	2.98*	1.47	0.96	1.45	1.26	
	Opportunity to catch big fish	−0.40	1.59	−0.23	2.18	1.64	0.99	
	Ability to keep caught fish	1.97*	2.67*	−1.08	2.06	1.50	2.01	
	The nature experience	−2.82**	1.57	0.26	0.95	0.32	1.88	
	Accessibility	0.01	2.53*	−2.50*	2.29	2.21	2.35	
	Being alone at the fishing spot	0.99	0.99	−0.71	0.48	0.91	1.08	
	Information on fishing rules	−0.03	1.42	−1.44	0.77	1.34	1.34	
	Socializing with family and friends	0.93	1.79	−1.26	1.33	1.50	1.75	
	Opportunity to relax	−1.47	0.87	−0.74	0.58	0.98	2.27	

Significance levels are indicated by asterisks at $p < 0.001$, $0.001 < p < 0.01$, and $0.01 < p < 0.05$.
[a]The test statistic Z is presented for the binary variables Study population, Gender, and Fished in the Archipelago.
[b]The test statistic maxT is presented for the ordinal variables Boat Use, Seasons, Residence, and Perceived Trend present for values from the one-way analysis with permutation test of independence.
[c]Seasons refers only to responses from Sportfiskarna members.

the general motivations of respondents that did or did not fish in the archipelago during the study period.

Discussion

Accurate, reliable human dimensions data is a key component of appropriate recreational fisheries management (Cooke et al. 2016; Holder et al. 2020; Hunt, Sutton, and Arlinghaus 2013). Yet errors and biases in survey data present a challenge to understanding social-ecological systems and achieving sustainable natural resource management. In the context of recreational fisheries, the investigation of survey respondent characteristics, their responses to human dimension and catch-related questions provide an important step toward making survey data more useful (Gundelund and Skov 2021; Holder et al. 2020). The present study examined the potential effects of various recruitment strategy proxies on different dimensions of recreational fisheries that are measured through surveys. All but one of the six recruitment strategies tested in this study revealed an effect on at least one of the psychological and/or catch-related human dimensions (Figure 3).

Several of the findings of this study are supported by other research looking into specific aspects of bias in angler surveys. Avidity, assessed by comparing members of Sportfiskarna to anglers in the general public, was found to influence responses to questions on catch-related and psychological dimensions of fishing, as hypothesized.

This result is in accordance with other studies that found differences in catch rates and motivations to fish (among other things) between high and low avidity anglers (e.g., Beardmore et al. 2011; van der Hammen, de Graaf, and Lyle 2016). Differences between the two initial sample frames were further emphasized by the fact that the response rate of the Sportfiskarna member sample was much higher than the response rate of the general public. These results illustrate how focusing data collection on groups of more avid anglers could generate results that cannot be extrapolated to the entire angling population. Conversely, surveys targeting only the general public have the potential to underrepresent perspectives, behaviors and trends that might be important for the management of specific fisheries that are targeted by more avid and specialized anglers (Gundelund et al. 2023; Lewin et al. 2023). It should be noted here that relationships between avidity and other human dimensions are highly variable and often fishery-specific (Hunt et al. 2023), suggesting fishery-specific influences of avidity on survey data.

The differences between genders found in this study suggest that missing responses from female fishers during data collections could lead to misinterpretation of catch motivations, and potentially overall harvest rates in recreational fisheries. Recreational angling is a male-dominated activity (Fennell and Birbeck 2019; Floyd et al. 2006), but this image may be wrongfully strengthened by the nonresponse of female anglers (Duda and Nobile 2010). The nonresponse error that leads to an overrepresentation of males among study participants is especially problematic in psychological surveys. The few studies that included enough female anglers to perform statistical analyses found that females are likely to differ in their attitudes and motivations to fish compared to male anglers (Schroeder et al. 2006; van den Heuvel et al. 2020). Moreover, with female participation in recreational fishing expected to increase in certain parts

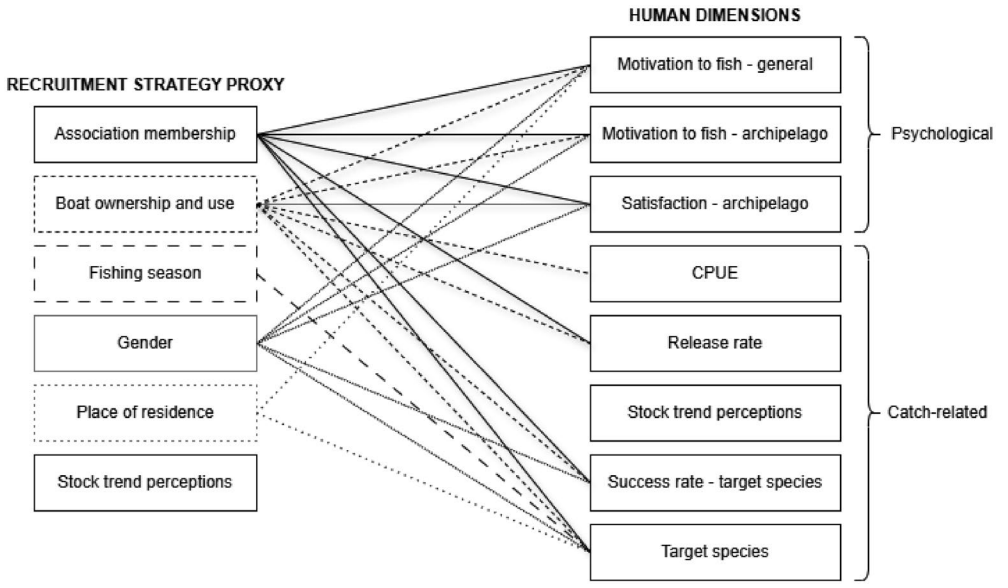


Figure 3. Effects of different types of recruitment strategy on the human dimensions included in this study.

of the world (Stensland et al. 2025), their representation in research becomes even more important. In this study, the significantly higher social motivation for fishing indicated by female respondents, as well as differences in score for the roles of nature experience and accessibility, provide insights into how differently angling is experienced and perceived by men and women. For some fishery stakeholders this could allude to where to focus additional attention if they are interested in recruitment and providing quality experiences that cater to more diverse groups of anglers.

Motivation and Satisfaction

Angler diversity is often simplified or categorized using indicators that are relevant for addressing particular fisheries management issues (e.g., more or less avid for effort and pressure on fisheries, catch orientation for harvest rates, and the conceptualizations of specialization and enduring involvement that may contribute to behavioral traits) (Hunt et al. 2023). However, in this study the responses linked to the general public and Sportfiskarna member samples (as respectively less and more avid anglers) are contrasted or contradicted by the proxies based on boat use and gender, contrary to our second hypothesis. Despite correlations across the boat use, gender, and public-Sportfiskarna sample frames, these characteristics yield different results for angler motivation and satisfaction, particularly with respect to catch-related satisfaction, accessibility, and social satisfaction (although these are not always significant). These findings demonstrate the importance of considering circumstances where particular defining characteristics may be more important than overall measures of avidity or specialization.

The results of this study also highlight several areas with notably limited differences in answers among potential recruitment strategies. Across all proxies the measures of satisfaction from fishing in the Roslagen archipelago appear to be relatively uniform, particularly compared to angler motivations. This suggests that satisfaction may be more universal and less prone to biases, while motivations are more strongly tied to particular angler characteristics and more vulnerable to biases. The lack of relationships between both seasonality and place of residence, and angler satisfaction and motivation suggest that these factors may be less important in research focused on human and psychological dimensions of recreational fisheries. This is not in agreement with previous studies that found differences between residents and nonresidents in attitudes and behaviors related to the fishery (Morales-Nin et al. 2015; Pokki et al. 2021). Moreover, in privately managed fisheries, residence may be connected to the ownership of fishing rights, which can influence the respondents' perspectives on recreational fishing activities and their management (Stensland 2013). Therefore, while the current study did not find differences in motivation or satisfaction, sampling only residents or only visitors of an area may result in error or bias for other human dimensions data. In addition, seasonality and the timing of data collection must still be considered due to recall biases that would shape both psychological and catch-related survey responses (Connelly, Brown, and Knuth 2000; Gundelund et al. 2023; Tarrant and Manfredo 1993). The scale of which can be rather large, as shown in Gundelund et al. (2023) where recall surveys demonstrate consistent 100–200% higher estimates of catches compared to data collected trip by trip.

Stock Trends and Target Species

Respondents' perceptions of stock trends were distinct as being relatively homogeneous across angler motivations, satisfaction, management factors, and target species, and no proxy effects were found (with the exception of boat use). The general perception from respondents was that stock trends are much worse, relative to the more positive responses provided in the previous study of this fishery (Soutukorva and Söderqvist 2005). It could have been expected that stock trend perceptions would influence angler behavior and psychological dimensions in some way, such as through lower retention rates, less satisfaction from catch, and different target species in response to their perceived presence or absence. The limited resources for fisheries management and conservation are often focused on mitigating negative stock trends for particular species (Ban et al. 2020; Bower et al. 2017; Cote et al. 2021), including research on the human dimensions of recreational fisheries (Holder et al. 2020; Hunt, Sutton, and Arlinghaus 2013). In this context the negative stock trend perception appears to be a unifying characteristic across angler groups in the Roslagen area and further investigation of this unity is warranted with respect to efforts to improve fish stocks in this area. These perceptions also agree with available studies for pike (Bergström et al. 2022), perch (Eklöf et al. 2020), and cod (Bryhn et al. 2022) suggesting that perceptions mirror reality in this case (See Supplementary Table 2 for breakdown of perceived trend by species).

The finding that nearly all recruitment strategy proxies had a significant relationship with the target species is relevant to both the data collection process and how that data can be applied in fisheries management. Fishing season and residence had significant differences in target species in the archipelago, despite a lack of influence on motivations, satisfaction, and release rates. This illustrates that although influences from season and residence could be dismissed based on the psychological dimensions and other catch-related factors in this study (as presented in Table 4), they must be considered important when focusing data collection on particular species. Several studies demonstrate that angler decisions and behavior are likely to differ depending on the targeted species (Beardmore et al. 2015; Birdsong et al. 2022; Tingley et al. 2019). Acknowledging the number of factors that influence target species is important to ensure that researchers can make accurate descriptions of angler behavior (Koemle et al. 2021), calculations of species-specific effort (Ochwada-Doyle et al. 2022), and trends such as target species substitution in multispecies fisheries (Ditton and Sutton 2004; Sutton and Ditton 2005). These findings also highlight opportunities to target data collection on angler groups that focus their fishing effort on species of interest, by such means as boat ramp surveys (Sande et al. 2022), or strategically timing surveys with respect to fishing seasons (Blyth and Rönnbäck 2022; van den Heuvel and Rönnbäck 2023).

Management Implications

Based on the findings of this study, we conclude that there is considerable potential for recruitment strategy to influence the outcomes and interpretation of natural resource user surveys, and angler surveys in particular. Considering how catch reports and

human dimensions data are used to inform recreational fisheries management, such influences could lead to mismatches that complicate the achievement of sustainable outcomes in resource management interventions. This is supported by other studies from the region that found sources of bias when extrapolating catch data from angling diaries (Lewin et al. 2023), and different angler survey platforms (Gundelund et al. 2023).

Several solutions have been raised to address error and biases in angler surveys, but they primarily focus on avidity and its effects on catch estimates. Some studies select a probability sample of the general public that should, theoretically, equalize the likelihood of being recruited into a study for avid and occasional anglers (NRC 2006; Roth et al. 2001). Others suggest that onsite intercept surveys should be used periodically to validate offsite surveys (Nieman et al. 2021; Lewin, Weltersbach, Haase, Riepe, et al. 2021). One drawback of such approaches is that a sampling error based on the spatial scale of the study might arise. In another attempt at correction for avidity, van der Hammen, de Graaf, and Lyle (2016) compared the effects of an avidity weighting procedure and a survey drop-out removal procedure to assess the effect of survey non-response on total catch estimates. However, similar weighting procedures for bias adjustment of psychological variables are currently unavailable. The development of such a procedure requires large amounts of data about the characteristics of both respondents and non-respondents, which is challenging to obtain (yet efforts are being made to help researchers overcome this barrier, see e.g., Biemer and Peytchev (2013)).

In addition to the development of weighting procedures, researchers and managers can also put effort into promoting the recruitment of underrepresented angler groups into angler surveys. Such efforts could include specific recruitment campaigns designed to attract female anglers or casual (i.e., low-avidity) anglers, for example through purposive sampling or snowball sampling variations, which have been found successful in other research fields (e.g., Sadler et al. (2010)). Further research focusing on the values and beliefs of these underrepresented angler groups would be necessary to design effective campaigns that align with their values.

We acknowledge that in some cases error or bias is either not a problem or not preventable. For example, there are a number of situations where anglers with certain characteristics are intentionally targeted because of their potentially pivotal role in a fishery (e.g., tournament anglers (Mannheim et al. 2018), anglers in fisheries with particularly high value or challenging target species (Blyth and Rönnbäck 2022; van den Heuvel et al. 2022), or fishing guides (Farthing et al. 2022). Representativeness is thus often relative to the aim of the study. In other cases, the resources (e.g., time or money) needed to overcome these challenges could be unavailable to managers or researchers. Ultimately, the chosen recruitment strategy will be the result of a tradeoff between the information needs and the resource availability. The results presented in this study help us to understand the implications of different recruitment strategies and as such, they may provide guidance to researchers and managers in the collection and interpretation of survey data.

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Disclosure statement

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